

Conflict-Aware Evaluation turns cross-view triage into a stress test

The method asks whether fusion resolves the cases where ground and overhead evidence disagree.

1 Paired evidence

overhead patch ground view



$$\mathcal{D} = \{(x_i^{(s)}, x_i^{(r)}, y_i)\}_{i=1}^N$$

PROPERTY-CENTRIC

target aligned, local cues

PANORAMIC

context rich, target weaker

Observation regime

observer-target geometry
changes which view arbitrates

2 Comparable predictors

$$E_s(x_i^{(s)})$$

$$p_i^{(s)}$$

$$E_r(x_i^{(r)})$$

$$p_i^{(r)}$$

late fusion

$$[z_s, z_r, |z_s - z_r|, z_s \odot z_r]$$

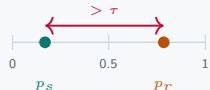
$$p_i^{(cv)}$$

same backbone, same split,
different evidence access

3 Conflict gate

$$\mathcal{C}_\tau = \{i : |p_i^{(s)} - p_i^{(r)}| > \tau\}$$

keep the hard cases
where views disagree



stress subset

evaluate models where
average F1 is least informative

threshold is tunable;
main analysis uses $\tau = 0.1$

4 CAE readout

overall F1
full test

conflict F1
on $\mathcal{C}_\tau$

Δ_{view}
CV minus best single view

Directional arbitration

S correct, R wrong

R correct, S wrong

switch reveals regime boundary

Report when fusion helps,
fails, or needs human review

Methodological claim: report conflict-subset behavior, not only average-case accuracy; CAE makes fusion falsifiable.